

Energy-Efficient Path Planning and Task Allocation for Multi-UAV Aided IoT Cluster-Based Data Collection

GUODONG ZHAO, Student Member, IEEE

JINJING WANG, Senior Member, IEEE

ZHIJUN MENG

ZICHEN WANG

HANG FU, Student Member, IEEE

Beihang University, Beijing, China

CHUNXIAO JIANG, Fellow, IEEE

Tsinghua University, Beijing, China

Abstract—In cluster-based Internet of Things (IoT) environments, unmanned aerial vehicles (UAVs) are increasingly being deployed for data collection tasks due to their agility and flexibility. However, the limited energy capacity of UAVs remains a significant challenge. In this paper, we address the joint optimization problem of task allocation and path planning for multi-UAV swarms, aiming to minimize energy consumption while ensuring safe operations in dynamic, real-time environments. We decompose the problem into sub-tasks including task allocation among multiple types of UAVs, and safe, energy-efficient path planning. To solve these sub-tasks, we introduce a centralized transformation of the traveling salesman problem (TSP) combined with a dynamic programming-based method for task allocation that integrates data transmission

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G. Zhao, Z. Meng and Z. Wang are with the School of Aeronautic Science and Engineering, Beihang University, Beijing 100191, China (e-mail: {zhaoguodong00, mengzhijun, wangzichen}@buaa.edu.cn). J. Wang and H. Fu are with the School of Cyber Science and Technology, Beihang University, Beijing 100191, China (e-mail: {drwangjj.zy2005116}@buaa.edu.cn). C. Jiang is with Beijing National Research Center for Information Science and Technology (BNRist), Tsinghua University, Beijing, 100084, China (email: jchx@tsinghua.edu.cn).

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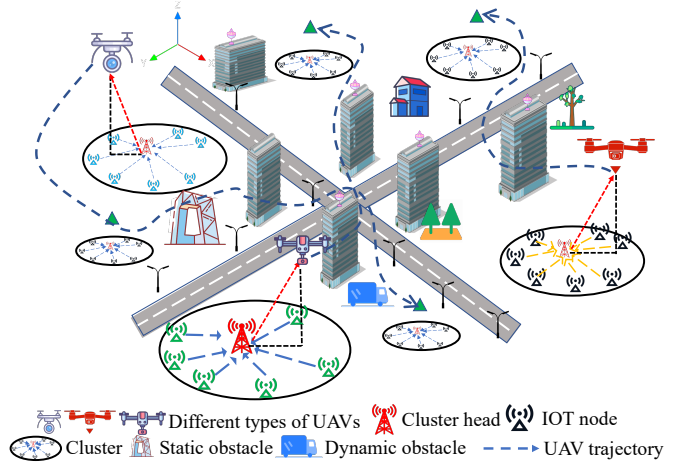


Fig. 1: The illustration of different UAVs in a smart city collection data scenario, where multi types of UAVs are equipped with different sensors corresponding to different tasks. The UAV management system performs task allocation and path finding based on the functions of UAVs and existing tasks.

considerations. For path planning and trajectory optimization, we leverage multi-agent deep reinforcement learning (RL) for safe way-point planning and design an energy-optimal planner to minimize energy consumption. Theoretical analysis and simulation results demonstrate that the proposed approach significantly improves energy efficiency compared to existing methods.

Index Terms—Multi-UAV, path planning, task allocation, energy optimization, Internet of Things (IoT)

I. Introduction

UNMANNED aerial vehicles (UAVs) have become a critical technology in smart cities and Internet of Things (IoT) applications due to their flexibility, cost-effectiveness, and ability to perform various tasks, including data collection, communication relays, and surveillance [1], [2]. UAVs can serve as communication intermediaries by hovering near the ground or objects, allowing them to collect data within their communication range [3], [4]. This capability reduces the communication load on ground nodes, making UAVs indispensable for a wide array of applications [5], [6].

However, the data collection process using UAVs introduces several challenges. A primary challenge is the increased energy consumption of UAVs, resulting from their need to frequently move and establish connections with multiple devices [2]. Additionally, in real-world environments, data-generating nodes are often distributed in clusters, necessitating a cluster-based data collection approach. In this approach, a unique cluster head within each cluster is responsible for communicating with the UAV, while other devices relay their data to the cluster head [7], [8], [9]. This cluster-based method reduces the communication burden on both UAVs and ground nodes by compressing data transmission and is widely adopted

in UAV-aided data collection systems [10]. In practical application scenarios, various types of IoT ground nodes are used for data collection, including surveillance cameras and air quality sensors. As these scenarios become increasingly complex and diverse, UAVs must be capable of performing distinct data collection functions tailored to specific tasks [11]. For example, in some scenarios, only a particular UAV—or the device it carries—is equipped to authenticate and establish a secure connection with an Internet node, thereby enabling effective communication with a cluster of IoT devices. This differentiation in data collection capability, rather than differences in sensor configuration alone, leads to the categorization of UAVs according to the tasks they can perform. Consequently, each scenario specifies a set of UAV types based on their unique data collection functions [12]. To address complex task scenarios and the limited endurance of individual UAVs, clusters of UAVs with varied data collection capabilities are typically deployed. These clusters comprise different UAV types in specific quantities to enhance task efficiency and ensure that each UAV is optimally assigned to the task that matches its data collection function.

When UAVs navigate between different hovering points, they may encounter dynamic or unmapped static obstacles. Therefore, UAVs require real-time obstacle avoidance capabilities [13]. Additionally, their flight trajectories should be as smooth as possible, avoiding sudden acceleration, deceleration, and sharp maneuvers to minimize energy consumption [14]. As a result, safe and energy-efficient motion planning for UAVs is crucial, necessitating real-time decision-making capabilities. Unlike task allocation in UAV clusters, this process requires UAVs to make dynamic decisions as they encounter obstacles during their flight.

A. Related Work

Numerous research efforts have been devoted to addressing the challenges associated with UAV-assisted data collection, particularly concerning energy consumption and path optimization. For instance, Ebrahimi et al. [7] proposed a method that clusters sensors for data collection, with UAVs subsequently collecting data from cluster heads. Similarly, Shen et al. [8] developed a UAV-supported data collection system for large-scale machine-type communication (mMTC), where machine-type communication devices (MTCs) are divided into clusters, and UAVs visit each hovering position corresponding to a cluster to collect data. Their approach focused on minimizing total system energy consumption by optimizing UAV trajectories.

Expanding on this, Zhu et al. [9] tackled the challenge of unknown cluster head positions, where the UAV's hovering points and visiting order are determined to minimize energy consumption. Further, Zhu et al. [10] formulated the problem of jointly optimizing hovering points and UAV trajectories as a TSP with neighborhood (TSPN), with the objective of minimizing the age of information

(AoI) in clustered IoT networks. These studies share a common goal of reducing UAV energy consumption through optimized data collection strategies [15]. However, they primarily focus on static environments and do not fully address dynamic obstacles or real-time decision-making.

To address these limitations, recent research has explored the use of heuristic algorithms and swarm intelligence optimization techniques in multi-UAV systems [16]. For example, particle swarm optimization (PSO) has been widely applied to solve path planning problems, including collision avoidance and coverage tasks in complex environments [17], [18], [19]. Although effective in certain scenarios, these approaches often assume static environments and are limited in their ability to handle dynamic obstacles [20], [21]. Additionally, multi-agent systems have been employed to address the pickup and delivery problem, where UAVs compute collision-free paths to execute tasks such as extraction and delivery [22], [23]. While these methods typically guarantee optimality through brute-force strategies, they struggle with scalability and performance in more dynamic settings [24].

Recent advancements suggest that learning-based approaches, particularly RL, can offer solutions to the limitations of traditional methods in dynamic environments. For example, RL has been applied to UAV pathfinding in dynamic environments, though these methods may not comprehensively address challenges such as non-cooperative obstacles and strict time constraints [25], [26]. Moreover, several studies have proposed joint trajectory and resource optimization frameworks to enhance the energy efficiency of UAV-assisted IoT systems [27], [28]. The integration of deep RL with multi-agent systems has shown promising results in optimizing both task allocation and trajectory planning [19]. LiDAR sensors used in our RL framework, with their high-precision 3D mapping and obstacle detection capabilities, are more effective for precise navigation than radar and ultrasonic sensors [29], but future sensor fusion may further enhance UAV performance.

Despite these advancements, challenges remain in ensuring scalability and solution quality in large-scale IoT networks where UAVs must operate under strict time constraints. Multi-UAV data collection strategies have focused on optimizing deployment and minimizing energy consumption, especially in scenarios with large numbers of IoT devices [17], [18], [21]. These strategies often emphasize collaboration among UAVs to improve efficiency and reduce energy consumption through cooperative behaviors [20], [27], [28]. Integrating various optimization techniques to manage the complexity of real-time multi-UAV coordination has emerged as a critical area of research [17].

B. Motivation and Contributions

In summary, while significant progress has been made in UAV-assisted data collection and path planning,

challenges related to real-time decision-making, dynamic obstacle avoidance, and energy efficiency remain. This paper seeks to address the aforementioned limitations by focusing on the path planning problem in UAV-assisted data collection, considering task allocation constraints. Specifically, the objective is to develop a comprehensive solution that not only ensures optimal energy consumption but also accounts for dynamic environmental factors and the need for real-time decision-making.

To achieve these goals, we propose a novel approach that decomposes the complex, non-convex optimization problem into several manageable sub-problems. These sub-problems include task allocation among UAVs with consideration of inter-cluster constraints, selection of hovering points within each cluster to optimize data collection efficiency, real-time safe path point selection for UAV movement between hovering points, and energy-optimal trajectory planning for UAVs to ensure minimal energy consumption during movement. We employ a centralized transformation of the TSP for task allocation and utilize a dynamic programming algorithm that integrates data transmission considerations to achieve optimal global assignment. After this centralized process, each UAV independently determines its own path planning and trajectory optimization. Specifically, a multi-agent deep RL framework is used onboard for safe path point planning, and an energy-optimal planner is implemented at the UAV level to minimize energy consumption.

The key contributions of this paper are summarized as follows:

- We establish an auxiliary IoT data collection model considering different types of UAV clusters, and formulate an optimization problem aimed at minimizing total energy consumption while accounting for both UAV data collection and path planning processes.
- We introduce an efficient problem decomposition strategy that divides the complex optimization problem into two sub-problems: task allocation and path planning. Each sub-problem is addressed using efficient algorithms, ensuring high-quality solutions.
- Through extensive simulation experiments, we demonstrate that our proposed approach effectively minimizes UAV energy consumption and outperforms existing methods in dynamic environments.

II. SYSTEM MODEL

The scenario depicted in Fig. 1 illustrates a multi-UAV task allocation model for efficient data collection in a densely populated urban environment with numerous IoT devices. In this context, UAVs are strategically assigned to various clusters of IoT nodes, which include smart infrastructure elements like traffic lights, environmental sensors, and surveillance cameras. The task allocation model optimizes the distribution of UAVs to ensure effective coverage of all clusters while minimizing energy consumption and avoiding collisions with static and dy-

namic obstacles. By determining the optimal sequence of cluster visits, the system ensures that each UAV operates within its endurance limits, enhancing data collection efficiency and reducing the overall energy footprint of UAV operations in this complex urban landscape.

A. UAV Data Collection Model

In this section, we introduce a model aimed at improving the data collection process using UAVs within an IoT ecosystem. The model operates in a 3D environment with dimensions $R \times R \times D$, optimized for the spatial distribution of M IoT nodes, which are grouped into M distinct clusters, denoted as $C = \{c_1, \dots, c_m, \dots, c_M\}$. Each cluster is managed by a strategically located cluster head, which is responsible for efficient data aggregation and communication. A UAV is assigned to collect data from each cluster.

More specifically, the positions of the cluster heads can be represented by the set V_{CH}

$$V_{CH} = \{ch_1, \dots, ch_m, \dots, ch_M\}, \quad (1)$$

where $ch_m = [x_{cm}, y_{cm}, 0] \in \mathbb{R}^3$, with x_{cm} and y_{cm} being the horizontal coordinates on the ground plane. The set of hovering points V_h is given by

$$V_h = \{h_1, \dots, h_m, \dots, h_M\}, \quad (2)$$

where $h_m = [x_{um}, y_{um}, z_{um}] \in \mathbb{R}^3$, with x_{um} and y_{um} as the horizontal coordinates and z_{um} denoting the flight altitude. Hence the euclidean distance between a hovering position and a cluster head node can be calculated by

$$d(ch_m, h_m) = \sqrt{(x_{um} - x_{cm})^2 + (y_{um} - y_{cm})^2 + z_{um}^2}. \quad (3)$$

The UAV data collection process involves two types of communication links, i.e. line of sight (LoS) and non-line of sight (NLoS). The probabilities of the LoS and NLoS are influenced by environmental factors such as the number and height of buildings in urban environments and the UAV's flight altitude, these probabilities are defined as [30]

$$P_{c_m}(LoS) = \frac{1}{1 + \alpha \exp(-\tilde{\alpha}(\theta_{c_m} - \alpha))}, \quad (4)$$

$$P_{c_m}(NLoS) = 1 - P_{c_m}(LoS), \quad (5)$$

respectively, where α and $\tilde{\alpha}$ are environment-dependent constants. And θ_{c_m} represents the elevation angle between the ground-based cluster head at position ch_m and the UAV at the hovering point h_m , and we have:

$$\theta_{c_m} = \arcsin\left(\frac{z_{um}}{d(ch_m, h_m)}\right). \quad (6)$$

The path loss between the UAV and the cluster head is described as [30]

$$L_{c_m}(LoS) = 20 \log_{10}(f_c d(ch_m, h_m)) + \xi(LoS), \quad (7)$$

$$L_{c_m}(NLoS) = 20 \log_{10}(f_c d(ch_m, h_m)) + \xi(NLoS), \quad (8)$$

where f_c is the carrier frequency, and $\xi(LoS)$ and $\xi(NLoS)$ represent additional path losses for LoS and

NLoS links, respectively, which are determined by the environment.

Average path loss between the cluster head ch_m and the hovering point h_m is given by

$$\bar{L}_{c_m} = P_{c_m}^{(LoS)} L_{c_m}^{(LoS)} + P_{c_m}^{(NLoS)} L_{c_m}^{(NLoS)}, \quad (9)$$

the average transmission rate r_{c_m} (in bits per second) between the cluster head ch_m and the UAV at hovering point h_m can be defined as

$$r_{c_m} = B_w \log_2(1 + SNR_{c_m}), \quad (10)$$

where B_w is the channel bandwidth and SNR_{c_m} is the signal-to-noise ratio (SNR), which is expressed as

$$SNR_{c_m} = \frac{P_{CH}}{\sigma^2 10^{\bar{L}_{c_m}/10}}. \quad (11)$$

Here, P_{CH} represents the transmission power of the cluster head, and σ^2 is the noise power at the UAV's receiver.

To ensure reliable data transmission, the distance between the UAV and the cluster head must satisfy the condition [31]

$$d(ch_m, h_m) < d_{ths}, \quad (12)$$

where d_{ths} denotes the maximum distance required to maintain reliable communication with the cluster head. Additionally, to ensure safety, it should satisfy

$$d_{safe} < d(ch_m, h_m), \quad (13)$$

where d_{safe} represents the minimum safe hovering distance.

B. UAV Energy Consumption Model

The energy consumption of a UAV generally consists of two primary components: communication-related energy and flying propulsion energy. The UAV energy consumption model accounts for both components to enable efficient data collection within an IoT framework.

As the UAV flies from one hovering point to another, it may frequently accelerate and decelerate depending on its instantaneous speed. For a rotary-wing UAV with instantaneous speed $V(t)$, the propulsion power consumption model can be expressed as [31]

$$P(V(t)) = P_0 \left(1 + \frac{3V^2}{U_{tip}^2} \right) + \frac{1}{2} d \rho A V^3 + P_1 \left(\sqrt{\sqrt{1 + \frac{V^4}{4v_0^4}} - \frac{V^2}{2v_0^2}} \right), \quad (14)$$

where P_0 and P_1 denote the constants representing blade profile power and induced power during hover, respectively, and U_{tip} denotes the tip speed of the rotor blade. Besides, v_0 denotes the mean rotor-induced velocity during hover, d is the UAV body drag ratio, while ρ and A denote the air density and rotor disk area, respectively.

When the UAV moves from the hovering point h_n to h_m over a total time t_{nm} , the propulsion energy consumed can be calculated as

$$E_{nm}^{fly} \{V(t), t_{nm}\} = \int_0^{t_{nm}} P(\|V(t)\|) dt. \quad (15)$$

During hovering at point ch_m , the hovering energy consumption E_{cm}^{hov} can be given by

$$E_{cm}^{hov} = T_{cm}^{hov} \cdot P(0), \quad (16)$$

where $P(0)$ is the power consumed during hovering and we have $P(0) = P_0 + P_1$.

The UAV's hovering time T_{cm}^{hov} at point ch_m , derived from the data collection process, includes both the data transmission time from regular nodes to the cluster head and the forwarding time from the cluster head to the UAV, i.e.

$$T_{cm}^{hov} = N_{bm} \tau + \frac{N_{bm} L_{data}}{r_{cm}}, \quad (17)$$

where N_{bm} represents the number of regular nodes surrounding cluster head b_m , while L_{data} and τ denote the data size per node and the time slot for transmission respectively.

The communication energy consumption E_m^{com} during data collection from the cluster head is

$$E_m^{com} = T_{cm}^{com} \cdot P_{com}, \quad (18)$$

where P_{com} denotes the constant power required for communication between the UAV and the cluster head, and T_{cm}^{com} specifies the communication duration.

Finally, the total energy consumption E_{total} of the UAV, accounting for both propulsion and communication energy, can be given by the sum of the propulsion energy during flight, the hovering energy, and the communication energy

$$E_{nm}^{total} = E_{nm}^{fly} \{q_{nm}(t), V(t), t_{nm}\} + E_{cm}^{hov} + E_m^{com}. \quad (19)$$

Although UAV energy is mainly consumed by propulsion and communication, the computational energy for autonomous navigation—determined by a fixed operating frequency—remains constant. Thus, over the flight time T , it equals $P_{comp} \cdot T$, adding only a constant to the objective and not affecting the optimal solution (see Appendix for detailed explanation).

C. Task Allocation Model for Multi-UAV Systems

In smart city applications, various types of IoT ground nodes, such as surveillance cameras and air quality sensors, are deployed in clusters for data collection. As these applications evolve, the complexity and diversity of data collection scenarios increase, necessitating the use of specific UAV types for efficient operations [12]. For example, in complex environments, UAVs equipped with high-quality LiDAR sensors may be required to perform tasks within certain clusters. However, not all clusters necessitate such specialized equipments, as it increases both economic costs and UAV payload. Therefore, each

cluster specifies the types of UAVs authorized to access and perform tasks in the area.

Each UAV begins with an initial battery charge, departs from a starting point, visits a series of clusters for data collection, and then flies to a designated destination to offload the collected data (which may also be the starting point).

To efficiently allocate tasks and plan routes for multiple UAVs, we propose a multi-UAV cluster sequence allocation model. Let $I = \{1, 2, \dots, N\}$ represent the set of UAV indices, where N is the total number of UAVs. The operational workspace is modeled as a graph $G = (V, E)$, where V denotes the set of vertices representing clusters, and E denotes the set of edges representing the connections between these clusters. The initial position of UAV i , corresponding to its starting point, is denoted by $v_i^o \in V$, and the set of all initial positions is represented as V_o . The set of destination points is denoted by $v_i^d \in V_d \subseteq V$, indicating possible landing points for UAVs. The set of task clusters is given by $V_c = V \setminus \{V_o \cup V_d\}$, representing the clusters where data collection tasks need to be performed. For each cluster c_i , the subset of UAVs eligible to visit c_i is denoted by $f_A(c_i) \subseteq I$.

The path of UAV i between clusters c_i^1 and c_i^ℓ is represented by a cluster sequence $\pi_i(c_i^1, c_i^\ell)$, which consists of a set of clusters $(c_i^1, c_i^2, c_i^3, \dots, c_i^\ell)$. The energy consumption for this path is denoted by $g(\pi_i(c_i^1, c_i^\ell))$, where

$$g(\pi_i(c_i^1, c_i^\ell)) = \sum_{\forall (v_i^p, v_i^q) \in \pi_i} E_{v_i^p v_i^q}^{\text{total}}. \quad (20)$$

The task allocation ensures that each cluster $v \in V_c$ is visited by at least one UAV from the set $f_A(v)$

$$\forall v \in V_c, \exists i \in f_A(v) \text{ such that } v \in \pi_i. \quad (21)$$

Each UAV $i \in I$ follows a unique path that starts at its initial vertex v_i^o and ends at a designated destination $u \in V_d$

$$\pi_i = (v_i^o, \dots, u), \text{ where } u \in V_d \text{ and } i \in f_A(u). \quad (22)$$

By satisfying these conditions, the task allocation model guarantees that all clusters are covered.

D. UAV Dynamics Models and Control Model

The dynamics of a UAV can be modeled using Newton's equations, which describe the relationship between forces, torques, and the resulting motion. For a typical quadrotor UAV, the dynamics can be given by [14]:

$$m\ddot{p} = -mg\mathbf{e}_3 + R\mathbf{F}_T, \quad (23)$$

where m and g denotes the mass of the UAV and the gravitational constant respectively, besides, the position of the UAV is given by $p = [x, y, z]^T$. And $\mathbf{e}_3 = [0, 0, 1]^T$ denotes the unit vector in the z-direction, while R denotes the rotation matrix from the body frame to the inertial frame. Finally, the total thrust generated by the propellers can be denoted as $\mathbf{F}_T$.

Next, we address the generation of motion primitives used in trajectory planning. The differential flatness property of quadrotor systems allows their trajectories to be represented by three independent one-dimensional time-parameterized polynomial functions [14]

$$p_\mu(t) = \sum_{j=0}^{P-1} \sum_{k=0}^K c_{jk} t^k \quad T_j \leq t < T_{j+1}, \quad (24)$$

where P denotes the the number of polynomial and K denotes the highest degree of each polynomial, $\mu \in \{x, y, z\}$, and $T_j \leq t < T_{j+1}$ represents the flight time allocated for each polynomial trajectory segment $[j, j+1]$. Quadrotor systems can be approximated as linear time-invariant (LTI) systems. Define the state vector as $\mathbf{x}(t) := [p(t)^T, \dot{p}(t)^T, \dots, p^{(n-1)}(t)^T]^T \in \mathcal{X} \subset \mathbb{R}^{3n}$. The control input is defined as $u(t) := p^{(n)}(t) \in \mathcal{U}$, where $v(t) \triangleq \dot{p}(t)$ represents the UAV velocity, and $a(t) \triangleq \ddot{p}(t)$ represents the UAV acceleration at time instant t .

The UAV must satisfy power system limitations given by the constraint to the max velocity

$$\|\dot{p}(t)\| \leq v_{\max}, \quad \forall t \in [0, T], \quad (25)$$

and the the constrain to the max acceleration

$$\|\ddot{p}(t)\| \leq a_{\max}, \quad \forall t \in [0, T], \quad (26)$$

Besides, safety constraints during flight must be followed

$$\|p(t) - p_{\text{obs}}(t)\| \geq d_{\text{safe}}, \quad \forall t \in [0, T]. \quad (27)$$

The state-space model can be defined as [32]

$$\dot{x} = \mathbf{A}x + \mathbf{B}u, \quad (28)$$

which describes the trajectory of the quadrotor system given an initial state $x(0)$ and a control input $u(t)$, where

$$\mathbf{A} = \begin{bmatrix} 0 & \mathbf{I}_3 & 0 & \cdots & 0 \\ 0 & 0 & \mathbf{I}_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \mathbf{I}_3 \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ \mathbf{I}_3 \end{bmatrix}. \quad (29)$$

During flight, based on the current state x , a set of discretized control signals u is input into the UAV's control system, and the UAV's state evolves according to Eq. (28) over time.

E. Problem Formulation for Multi-UAV Energy Minimization

In this section, we formulate the problem of minimizing energy consumption for a fleet of UAVs, while ensuring that all designated task points are completed. The primary objective is to minimize the total energy consumed by all UAVs during their missions. This can be achieved by optimizing the sum of the energy consumed along the paths taken by each UAV, which is represented by the function $g(\pi_i(v_i^1, v_i^\ell))$. Therefore, the objective

function is defined as follows

$$\begin{aligned}
& \underset{\{\pi, p, u\}}{\operatorname{argmin}} \sum_{i=1}^N g(\pi_i(v_i^1, v_i^{\ell})) \\
\text{s.t. } & \text{C1 : } d_{\text{safe}} < d(ch_m, h_m), \quad \forall m \in [0, M], \\
& \text{C2 : } \|p(t) - p_{\text{obs}}(t)\| \geq d_{\text{safe}}, \quad \forall t \in [0, T], \\
& \text{C3 : } \|\dot{p}(t)\| \leq v_{\text{max}}, \quad \forall t \in [0, T], \\
& \text{C4 : } \|\ddot{p}(t)\| \leq a_{\text{max}}, \quad \forall t \in [0, T], \\
& \text{C5 : } T_{cm}^{\text{hov}} \geq N_{bm}\tau + N_{bm}L_{\text{data}}/r_{cm}, \quad \forall m \in [0, M], \\
& \text{C6 : } v_i^1 == v_i^o, v_i^{\ell} == v_i^d, \quad \forall i \in [0, N], \\
& \text{C7 : } v \in \pi_i, \quad \forall v \in V_c, \exists i \in f_A(v), \\
& \text{C8 : } T_j \leq t < T_{j+1}, \quad \forall j \in [0, P], \\
& \text{C9 : } d(ch_m, h_m) < d_{\text{ths}}, \quad \forall m \in [0, M].
\end{aligned} \tag{30}$$

To explain further, constraints C1 and C2 are safety constraints expressed in (13) and (27), while dynamic constraints for UAVs expressed in (25) and (26) can be denoted as C3 and C4. Besides, the hovering time constraint C5 derived from (17) must be considered in optimization process. C6 denotes path continuity constraint derived from (22) while C7 denotes task completion constraint derived from (21). Trajectory time allocation constraint C8 expressed in (24) and distance constraint C9 for data transmission expressed in (12) also need to be met.

III. Problem Analysis and Decomposition

The problem of minimizing the total energy consumption for all UAVs, as described in (30), is inherently complex due to the non-linear nature of the objective function and constraints. It involves multiple intertwined aspects, including task allocation, the selection of optimal hovering points, path planning considering static and dynamic obstacles, and trajectory optimization for energy minimization. To effectively tackle this problem, it is essential to decompose it into more manageable sub-problems, each addressing a specific aspect of the overall challenge.

REMARK 1 *The energy consumed during UAV flight is significantly greater than the energy consumed during data collection, including both hovering and communication energy. Mathematically, for any cluster $c_m \in C$, this relationship can be expressed as*

$$\int_0^{t_{nm}} P(\|v(t)\|) dt \gg T_{\text{hov}} \cdot P(0) + T_{\text{com}} \cdot P_{\text{com}}. \tag{31}$$

A. Cluster Sequence Allocation and Hovering Point Selection

Given the significant impact of flight energy on total energy consumption, our first step is to optimize the sequence in which UAVs visit the clusters and select the hovering points within each cluster. The problem can be decomposed into two key sub-problems: (1) determining the optimal sequence of clusters that each UAV should

visit, and (2) selecting the optimal hovering points within each cluster to balance flight and hovering energy consumption.

1. Cluster Sequence Allocation

The sequence in which UAVs visit clusters should be optimized to minimize the total distance traveled. Based on Remark 1, and considering that the UAVs hover directly above the cluster head nodes, the problem can be formulated as follows

$$\begin{aligned}
& \underset{\pi}{\operatorname{argmin}} \sum_{i=1}^N \sum_{\forall (v_i^p, v_i^q) \in \pi_i} L(v_i^p, v_i^q) \\
\text{s.t. } & \text{C6, C7.}
\end{aligned} \tag{32}$$

Here, $L(v_i^p, v_i^q)$ denotes the distance between any two clusters v_i^p and v_i^q in the sequence π_i . Since the distance is based on the cluster head nodes, this problem assumes that UAVs hover directly above these nodes during data collection.

REMARK 2 *The solution space of the cluster sequence allocation problem (32) is a subset of the solution space of the original energy minimization problem (30). This is because the allocation problem focuses on minimizing flight distance, which dominates the overall energy consumption. Therefore, optimizing the cluster sequence ensures that the solution contributes positively to minimizing the total energy consumption.*

2. Hovering Point Selection

Once the cluster sequence is determined, the next step is to fine-tune the selection of hovering points within each cluster. The goal is to minimize the total energy consumption, including both flight and hovering energy. While the cluster sequence assumes that UAVs hover directly above the cluster head nodes, this sub-problem considers adjustments to these positions to achieve better energy efficiency. The hovering point selection problem can be formulated as

$$\begin{aligned}
& \underset{\{h_m^{(i)}\}}{\operatorname{argmin}} \sum_{i=1}^N \left(\sum_{c_m \in \pi_i} E_{cm}^{\text{hov}} + \sum_{(c_m, c_{m+1}) \in \pi_i} E_{cm, c_{m+1}}^{\text{fly}} \right) \\
\text{s.t. } & \text{C1, C5, C9.}
\end{aligned} \tag{33}$$

REMARK 3 *The selection of optimal hovering points, as formulated in (33), refines the solution obtained from the cluster sequence allocation. By optimizing both flight and hovering energy, the solution space of this problem extends the solution space of the original problem, meaning that solving this sub-problem brings the overall solution closer to the global optimum.*

B. Path Optimization and Trajectory Planning

After determining the optimal cluster sequence and hovering points, the final stage involves planning the

UAVs' paths between these hovering points and optimizing their trajectories to ensure minimal energy consumption during flight.

1. Safety Path Planning

The UAVs need to navigate safely from one hovering point to another while avoiding obstacles. The path planning problem can be formulated as

$$\begin{aligned} & \underset{P_{mm+1}^{(i)}}{\operatorname{argmin}} (L(h_m^{(i)}, u_{m+1}^{(i)})) \\ \text{s.t. } & \text{C2.} \end{aligned} \quad (34)$$

The objective is to minimize the distance $L(h_m^{(i)}, u_{m+1}^{(i)})$ between safe path points, ensuring that the UAVs avoid obstacles while minimizing energy consumption.

2. Trajectory Optimizing

Finally, once the safe path points are determined, the UAV must follow this path with minimal energy consumption, we have

$$\text{C10: } p(t_j) = P_{mm+1}^{(i)}(k), \quad \forall k \in [0, l], \exists t_j \in [0, t_{mm+1}]. \quad (35)$$

So the trajectory optimization problem can be formulated as

$$\begin{aligned} & \underset{u(t)}{\operatorname{argmin}} (E_{c_m c_{m+1}}^{fly}(u(t))) \\ \text{s.t. } & \text{C3, C4, C8, C10.} \end{aligned} \quad (36)$$

This optimization ensures that the UAV follows the path with smooth velocity and acceleration profiles, thereby minimizing energy consumption while adhering to dynamic constraints.

REMARK 4 *Each sub-problem's solution is critical to solving the original problem. By optimizing the cluster sequence, hovering points, and flight paths, we ensure that the UAVs operate with minimal energy consumption. The sequential optimization of these sub-problems aligns with the overall objective of minimizing total energy, making the proposed decomposition approach both effective and efficient.*

IV. ALGORITHM DESIGN

A. Transformation Method for Multi-UAV Cluster Sequencing

The cluster sequence allocation problem (32) for multi-UAV systems is closely related to the multi-agent combinatorial path planning (MCPF) problem [33], but it does not require resolving path conflicts between UAVs. To tackle this challenge, we propose a novel approach by transforming the problem into a multi-traveling salesman problem (mTSP). In our framework, the task allocation is conducted in a centralized manner at the base station, ensuring that the optimal cluster sequence is determined from a global perspective before being assigned to each UAV. After this centralized task assignment, individual UAVs independently perform local path planning and

trajectory optimization onboard. The key idea behind our method is as follows: First, the original graph of cluster nodes and allocation constraints is reformulated as a single TSP. In this transformation, the starting point is assigned to the first UAV, and the endpoint is designated for the N-th UAV. By solving the resulting TSP using an existing solver, the solution is then mapped back to derive the optimal cluster allocation for the multi-UAV scenario. The step-by-step process is outlined in Algorithm 1.

Algorithm 1 Transformation of mTSP for Multi-UAV Cluster Allocation

Input: $G = (V, E, C)$ ▷ Target graph
Input: f_A ▷ Assignment constraints
Output: π^* ▷ Optimal joint sequence

```

1: for all  $v \in V_c \cup V_d$  do
2:   for all  $i \in f_A(v)$  do
3:      $U[v].\text{add}(v^i)$ 
4:   end for
5: end for
6:  $(V_{TF}, E_{TF}, C_{TF}) \leftarrow (V \cup U, \{\}, \{\})$ 
7: for all  $v \in V_c \cup V_d$  do
8:   for all  $i \in f_A(v)$  do
9:      $E_{TF}.\text{add}((v^i, \text{Next}(v^i)), 0)$ 
10:    for all  $u \in U[v]$  do
11:       $E_{TF}.\text{add}((v_i^o, u), C(v_i^o, u))$ 
12:      if  $u \neq v^i$  then
13:         $E_{TF}.\text{add}((\text{Prev}(v^i), u), Z + C(v, u))$ 
14:      end if
15:    end for
16:   end for
17: end for
18: for all  $i \in \{1, 2, \dots, N-1\}$  do
19:    $E_{TF}.\text{add}((v_i^o, v_{i+1}^d), 0)$ 
20: end for
21:  $E_{TF}.\text{add}((v_N^d, v_1^o), 0)$ 
22:  $\tau^* \leftarrow \text{TSP\_Solver}(V_{TF}, E_{TF}, C_{TF})$ 
23:  $\pi^* \leftarrow \{\}$ 
24: for all segment  $\in \tau^*$  do
25:    $\pi^*.\text{append}(\text{segment})$ 
26: end for
27: return  $\pi^*$ 

```

Algorithm 1 begins by initializing a set $U[v]$ for each vertex v in the clusters C and destination vertices V_d . For each vertex v , and for each UAV i assigned to v according to the constraints f_A , a duplicate vertex v^i is created and added to $U[v]$ (lines 1–5). This ensures that each cluster and destination vertex is represented for all UAVs eligible to visit them, preserving the assignment constraints in the new transformed graph (V_{TF}, E_{TF}, C_{TF}) (line 6).

Next, for each vertex v in $C \cup V_d$, and for each UAV i in $f_A(v)$, an edge from v^i to the next vertex $\text{Next}(v^i)$ with zero cost is added to E_{TF} (line 9). Additionally, for each pair of augmented vertices u in $U[v]$, an edge is added between v^i and u with cost $C(v^i, u)$ (lines 10–11). An edge from the previous vertex $\text{Prev}(v^i)$ to u is added with a cost of $Z + C(v, u)$ if $u \neq v^i$ (lines 12–13), where Z

is a large constant. These edges ensure high-cost penalties for shortcuts, maintaining the sequence constraints.

To maintain continuity in the path sequences across UAVs, an edge is added from the initial vertex v_i^o to the corresponding destination vertex v_{i+1}^d with zero cost for each UAV i from 1 to $N - 1$ (lines 18-20). Additionally, an edge from the destination vertex of the last UAV v_N^d to the initial vertex of the first UAV v_1^o is added with zero cost (line 21). The transformed graph (V_{TF}, E_{TF}, C_{TF}) is then solved using a TSP solver to find the optimal tour τ^* (line 22). The optimal sequence π^* is then generated by appending each segment of τ^* (lines 24-26), ultimately forming the optimal cluster joint sequence π^* (lines 25).

Algorithm 1 reduces the multi-UAV cluster allocation problem to a single TSP, leveraging established TSP-solving techniques to achieve an optimal or near-optimal solution.

B. Dynamic Programming for Hovering Points Selection

Algorithm 2 Dynamic Programming for UAV Hovering Points Selection

Input: $V_{CH} = \{CH_1, CH_2, \dots, CH_M\}$ $\triangleright$ Cluster heads
Input: V_H $\triangleright$ All possible hovering points
Input: $f_{CH \rightarrow H}$ $\triangleright$ Mapping from V_{CH} to V_H
Output: Π^* $\triangleright$ Optimal path with minimum energy

- 1: $s \leftarrow CH_1, g \leftarrow CH_M, N \leftarrow$ number of clusters
- 2: Initialize DP and $Prev$ as empty tables
- 3: **for** each H_j in $f_{CH \rightarrow H}(CH_2)$ **do**
- 4: $DP(1, H_j) \leftarrow E_{move}(s, H_j, CH_2)$
- 5: $Prev(1, H_j) \leftarrow s$
- 6: **end for**
- 7: **for** $i \leftarrow 2$ to N **do**
- 8: **for** each H_j in $f_{CH \rightarrow H}(CH_i)$ **do**
- 9: $E_{opt} \leftarrow \infty, H_{prev} \leftarrow \text{None}$
- 10: **for** each H_k in $f_{CH \rightarrow H}(CH_{i-1})$ **do**
- 11: $E_{total} \leftarrow DP(i-1, H_k) + E_{move}(H_k, H_j, CH_i)$
- 12: **if** $E_{total} < E_{opt}$ **then**
- 13: $E_{opt} \leftarrow E_{total}, H_{prev} \leftarrow H_k$
- 14: **end if**
- 15: **end for**
- 16: $DP(i, H_j) \leftarrow E_{opt}, Prev(i, H_j) \leftarrow H_{prev}$
- 17: **end for**
- 18: **end for**
- 19: Initialize Π^* as an empty list
- 20: $H_{current} \leftarrow g$
- 21: **for** $i \leftarrow N$ down to 1 **do**
- 22: Append $H_{current}$ to Π^*
- 23: $H_{current} \leftarrow Prev(i, H_{current})$
- 24: **end for**
- 25: Append s to Π^*
- 26: **return** $\Pi^*, DP(N, H_{current})$

Building on the cluster sequence obtained from Algorithm 1, a dynamic programming algorithm is proposed to

select the optimal hovering points for each UAV within each cluster, addressing problem (33). The goal of the algorithm is to minimize overall energy consumption, including both flight and hovering energy.

The hovering area within each cluster is divided into cubic grids, each grid representing a potential hovering point. Let $V_H = \{H_1, H_2, \dots, H_k\}$ denote the set of all possible hovering points for all clusters, which satisfies the safety constraint (13) and the distance constraint (12) for data transmission. Let $DP(i, H_j)$ represent the minimum energy required to reach hovering point H_j in cluster i . The optimal substructure property of this dynamic programming algorithm can be expressed as

$$DP(i, H_j) = \min_{H_k \in f_{CH \rightarrow H}(CH_{i-1})} (DP(i-1, H_k) + E_{move}(H_k, H_j, CH_i)), \quad (37)$$

where $DP(i, H_j)$ represents the minimum energy required to reach hovering point H_j in cluster i , $DP(i-1, H_k)$ is the minimum energy required to reach hovering point H_k in cluster $(i-1)$, $E_{move}(H_k, H_j, CH_i)$ denotes the energy required to move from hovering point H_k in cluster $(i-1)$ to hovering point H_j in cluster i , including both flight and hovering energy, $f_{CH \rightarrow H}(CH_{i-1})$ represents all possible hovering points in cluster $(i-1)$. The final solution is given by

$$\min_{H_j \in f_{CH \rightarrow H}(CH_N)} (DP(N, H_j) + E_{move}(H_j, g, CH_N)). \quad (38)$$

The dynamic programming process in Algorithm 2 begins by initializing the start node s as the first cluster head CH_1 and the goal node g as the last cluster head CH_M (line 1). The algorithm proceeds by initializing the energy table DP and the predecessor table $Prev$ to store intermediate results (line 2). Next, the algorithm calculates the initial energy required to move from the start node s to each possible hovering point in the second cluster, storing these values in the DP table, and records the start node as the predecessor (lines 3-5).

The main loop then iterates over all clusters from the second cluster to the last (lines 7-18). For each cluster, the algorithm examines all possible hovering points H_j within the cluster. It computes the total energy E_{total} required to move from any hovering point H_k in the previous cluster to H_j and adds it to the minimum energy required to reach H_k . The algorithm selects the path with the minimum energy and updates DP and $Prev$ accordingly (lines 8-16). After processing all clusters, the algorithm reconstructs the optimal path Π^* by backtracking from the goal node g using the predecessor table $Prev$. The backtracking process continues until the start node s is reached (lines 21-24). The final path Π^* and its corresponding minimum energy $DP(N, H_{current})$ are then returned (line 26).

C. RL-based Algorithm for Safety Waypoint Planning

After determining the hovering points for all clusters, the Proximal Policy Optimization (PPO) algorithm

is employed to train a neural network using a reward function that comprehensively considers both path safety and energy consumption. The trained network is then deployed onboard each UAV to identify safe waypoint paths, ensuring navigation from one hovering point to the next while avoiding both dynamic and static obstacles. This approach effectively addresses the problem defined in Eq. (34).

The state space $\mathcal{S}$ for each UAV consists of intrinsic and extrinsic components. The intrinsic state $\mathbf{s}_{\text{intr}}$ includes the UAV's position $\mathbf{p} \in \mathbb{R}^3$, velocity $\mathbf{v} \in \mathbb{R}^3$, and orientation $\mathbf{q} = \{q_w, q_x, q_y, q_z\} \in \mathbb{R}^4$. The extrinsic state is captured via lidar sensors, represented as $\mathbf{s}_{\text{lidar}} \in \mathbb{R}^{h \times w}$, where h and w denote the resolution of the lidar scan. The full state space is thus expressed as

$$\mathcal{S} = \{\mathbf{s}_{\text{intr}}, \mathbf{s}_{\text{lidar}}\}. \quad (39)$$

The action space $\mathcal{A}$ involves selecting a safe waypoint $\mathbf{a} \in \mathbb{R}^3$ for each UAV to guide it through the environment.

The reward function $r(\mathbf{s}, \mathbf{a})$ is designed to encourage safe and energy-efficient navigation. The total reward is calculated as the sum of multiple components

$$r(\mathbf{q}, \mathbf{s}, \mathbf{a}) = \alpha_1 r_{\text{vel}}(\mathbf{s}, \mathbf{a}) + \alpha_2 r_{\text{safety}}(\mathbf{s}) + \alpha_3 r_{\text{effort}}(\mathbf{a}) + \alpha_4 r_{\text{action}}(\mathbf{a}, \mathbf{a}_{\text{prev}}). \quad (40)$$

Each term in the reward function is defined as follows:

- Velocity reward: Encourages efficient movement towards the target:

$$r_{\text{vel}}(\mathbf{s}, \mathbf{a}) = \mathbf{v} \cdot \frac{\mathbf{p}_{\text{target}} - \mathbf{p}}{\|\mathbf{p}_{\text{target}} - \mathbf{p}\|}, \quad (41)$$

where $\mathbf{p}_{\text{target}} \in \mathbb{R}^3$ is the target position in 3D space and $\mathbf{p}$ is the current position of the UAV. This term encourages movement towards the target by maximizing the projection of the velocity vector $\mathbf{v}$ onto the direction vector $\mathbf{p}_{\text{target}} - \mathbf{p}$.

- Safety reward: Penalizes proximity to obstacles based on lidar data:

$$r_{\text{safety}}(\mathbf{s}) = \log(d_{\text{obs}}(\mathbf{s}) - d_{\text{safe}}), \quad (42)$$

where $d_{\text{obs}}(\mathbf{s})$ represents the distance to the nearest obstacle based on lidar readings.

- Effort reward: Penalizes high energy consumption:

$$r_{\text{effort}}(\mathbf{a}) = -E_{\text{fly}}(\mathbf{a}), \quad (43)$$

where $E_{\text{fly}}(\mathbf{a})$ denotes the energy required for the action $\mathbf{a}$. This term discourages actions that require high energy consumption.

- Action smoothness reward: Encourages smoother waypoints by penalizing significant changes in action:

$$r_{\text{action}}(\mathbf{a}, \mathbf{a}_{\text{prev}}) = -\|\mathbf{a} - \mathbf{a}_{\text{prev}}\|^2, \quad (44)$$

where $\mathbf{a}_{\text{prev}}$ denotes the previous action. This term penalizes large deviations from the previous waypoint, promoting smooth transitions.

The objective of the PPO algorithm is to find the optimal policy π^* that maximizes the expected cumulative

reward

$$\pi^* = \underset{\pi}{\operatorname{argmax}} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r(\mathbf{s}_t, \mathbf{a}_t) \right], \quad (45)$$

where the discount factor $\gamma \in [0, 1]$ controls the importance of future rewards. The PPO algorithm optimizes the policy using a clipped surrogate objective

$$L_t^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\mathcal{R}_t(\theta) \hat{A}_t, \text{clip}(\mathcal{R}_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (46)$$

where ϵ is the clipping parameter and $\mathcal{R}_t(\theta)$ denotes the probability ratio

$$\mathcal{R}_t(\theta) = \frac{\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_{\theta_{\text{old}}}(\mathbf{a}_t | \mathbf{s}_t)}. \quad (47)$$

The advantage function $\hat{A}_t$ is computed using Generalized Advantage Estimation (GAE), and the value function $V_{\phi}(\mathbf{s}_t)$ is updated by minimizing the following loss function

$$L^{\text{VF}}(\phi) = \mathbb{E}_t \left[\left(V_{\phi}(\mathbf{s}_t) - \hat{R}_t \right)^2 \right]. \quad (48)$$

Through this process, the UAVs learn to navigate safely through complex environments while minimizing energy consumption, ultimately converging on an optimal set of waypoints for each mission.

D. Trajectory Point Tracking Flight with Minimum Energy Control

This section discusses trajectory point tracking flight with minimum energy control. Our approach utilizes geometrically constrained trajectory optimization to generate trajectories that are both dynamically feasible and minimize control effort. The methodology is extended to address the energy optimization problem (36) in UAV trajectory optimization.

First, the system dynamics performance constraints of the UAV are incorporated into the objective function as penalty terms, transforming the problem (36) into

$$\underset{u(t)}{\operatorname{argmin}} (E_{c_m c_{m+1}}^{\text{fly}}(u(t)) + w_v \rho_v + w_a \rho_a) \quad (49)$$

s.t. C8, C10,

where w_v and w_a are corresponding parameters and

$$\begin{aligned} \rho_v &= \sum_{j=1}^M \int_{T_{j-1}}^{T_j} (\|p_j^{(1)}(t)\| - v_{\text{max}})^2 dt, \\ \rho_a &= \sum_{j=1}^M \int_{T_{j-1}}^{T_j} (\|p_j^{(2)}(t)\| - a_{\text{max}})^2 dt. \end{aligned} \quad (50)$$

According to Eq. (24), the objective function is equivalent to a function $H(C, T)$ that takes the polynomial parameters C and the time variable T as inputs, which can be solved by the MINCO (Minimum Control) trajectory parameterization method proposed in [14]. The MINCO minimizes control effort by using sparse parameters to

TABLE I: Simulation and Training Parameters

Parameter	Value	Parameter	Value
β	0.11	β	12.08
δ	0.5	P_{CH}	0.1 W
γ_{th}	20 dB	ξ_{LoS}	1 dB
ξ_{NLoS}	20 dB	σ^2	-110 dBm
γ	0.99	f_c	2 GHz
L_{data}	5 Mb	B_w	1 MHz
P_{com}	0.1 W	L_{sub}	5
P_0	99.66 W	P_1	120.16 W
U_{tip}	120 m/s	v_0	0.002 m/s
d	0.48	ρ	1.225 kg/m ³
ϵ	0.01	d_{safe}	10m
w_v	1000	w_a	1000
v_{max}	15 m/s	a_{max}	3 m ³ /s

control both the spatial and temporal profiles of the trajectory. The trajectory class can be defined as follows

$$\mathcal{T}_{MINCO} = \{p(t) : [0, T] \rightarrow \mathbb{R}^m \mid \mathbf{c} = \mathbf{c}(\mathbf{q}, \mathbf{T}) \mid \forall \mathbf{q} \in \mathbb{R}^{m \times (M-1)}, \mathbf{T} \in \mathbb{R}_{>0}^M\}. \quad (51)$$

where $\mathbf{q}$ represents the trajectory points at each instant t , and $\mathbf{T}$ is the time vector.

According to the definition and mapping method of the optimal MINCO trajectory in the MINCO framework, we have

$$b(q) = MC, \quad (52)$$

where $b(q)$ is a vector related to the trajectory points defined in MINCO, and M is a non-singular mapping matrix. Thus, our objective function can be transformed into a function of the path point $\mathbf{q}$ and the time variable $\mathbf{T}$ as follows

$$\underset{u(t)}{\operatorname{argmin}} \hat{H}(\mathbf{q}, \mathbf{T}) = H(C(\mathbf{q}, \mathbf{T}), \mathbf{T}) \quad (53)$$

s.t. C8, C10.

Furthermore, the gradients with respect to $\mathbf{q}$ and $\mathbf{T}$ are as follows

$$\frac{\partial \hat{H}(\mathbf{q}, \mathbf{T})}{\partial \mathbf{q}} = \frac{\partial H}{\partial \mathbf{C}} \frac{\partial \mathbf{C}}{\partial \mathbf{q}}, \quad (54)$$

$$\frac{\partial \hat{H}(\mathbf{q}, \mathbf{T})}{\partial \mathbf{T}} = \frac{\partial H}{\partial \mathbf{T}} + \frac{\partial H}{\partial \mathbf{C}} \frac{\partial \mathbf{C}}{\partial \mathbf{T}}, \quad (55)$$

where $\frac{\partial \mathbf{C}}{\partial \mathbf{q}}$ and $\frac{\partial \mathbf{C}}{\partial \mathbf{T}}$ can be computed in a linear-complexity manner by leveraging the results in [14]. The method proposed in [14] is then used to eliminate the temporal constraints (24) and spatial constraints (35). Consequently, the constrained optimization problem is transformed into an unconstrained one that can be efficiently and robustly solved using quasi-Newton methods [34].

V. SIMULATION RESULTS

A. Simulation Setting

The computational framework was implemented in Python, with the multi-UAV environment developed using Nvidia IsaacSim. Neural networks were constructed and

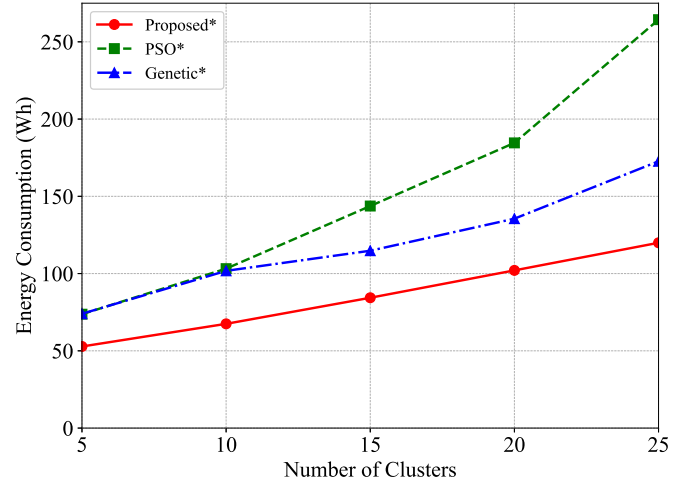


Fig. 2: Total energy consumption for different algorithms

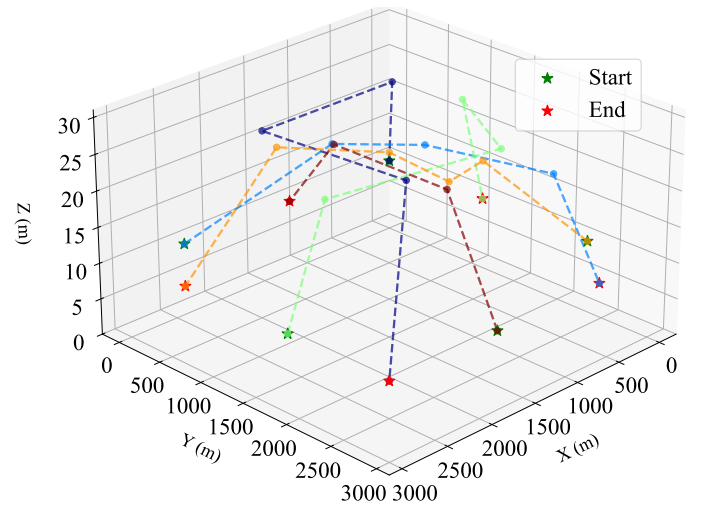


Fig. 3: Example of hovering point visit sequence for five UAVs after task allocation and hovering point selection to collect data from 15 clusters.

trained with PyTorch on an Ubuntu 20.04 system (i9-13900KF CPU, 64 GB RAM, NVIDIA RTX 4080 GPU). The simulation and training parameters are detailed in Table I. A supplementary introduction and video demonstrating the simulation process can be found online¹.

B. Analysis of Total Energy Consumption

To evaluate the total energy consumption of the proposed algorithm framework for multi-UAV data collection, referring to [10], we performed a comparative analysis with heuristic-based algorithms, specifically the Genetic Algorithm [35] and the PSO algorithm [36], both used for selecting hovering points for multiple UAVs. The parameters for these heuristic algorithms were meticulously adjusted to ensure optimal performance. The

¹<https://github.com/BUAA-SIC-LAB/EEPTA-MultiUAV>

simulation tests assumed a static environment devoid of dynamic obstacles, rendering it suitable for the application of the existing heuristic algorithms. The experimental setup utilized a $3km \times 3km \times 200m$ map, with five different types of UAVs operating over randomly generated clusters, each containing 15 IoT nodes. The set of UAVs $f_A(c_i)$ that could access each cluster c_i was also randomly generated. Multiple experiments were conducted, and the results were averaged.

Fig. 3 shows our algorithm consistently achieves lower energy consumption than PSO and Genetic algorithms across 5-25 clusters. At 5 clusters, our method uses only 52.81 Wh versus 73.71 Wh (PSO) and 73.97 Wh (Genetic). At 25 clusters, it requires 119.93 Wh compared to 264.34 Wh (PSO) and 172.67 Wh (Genetic), saving 54.6% and 30.5% respectively. Energy consumption increases for all algorithms as cluster count grows, with ours maintaining the lowest rate of increase. This superior performance is attributed to the high solution quality of the algorithms used in our framework for solving the hovering point sequence selection and assignment problems, including the multi-traveling salesman transformation algorithm and the obstaclesprogramming hovering point selection algorithm, which effectively minimize energy expenditure across varying cluster densities.

C. RL-based Safety Waypoint Planning

In this paper, we propose a RL-based algorithm for generating safe waypoints to navigate in environments with unknown static and dynamic obstacles. The training process consists of a total of 150 million steps, with 1024 UAVs operating in parallel training environments. Each UAV is allowed a maximum of 800 steps per training episode. Obstacles are randomly generated, and the UAV's starting point and destination are set 20 meters apart in a straight line. The weighting of factors of reward function are [1.0, 0.2, 0.1, 1]. The actor network in the PPO algorithm is divided into an encoder layer and a decision layer. The encoder layer, primarily responsible for processing state information such as radar data, is composed of multiple convolutional neural network layers. The decision layer consists of a Long Short-Term Memory (LSTM) network, while the critic network is built using a Multi-Layer Perceptron (MLP) network. Through parallel training, the UAVs learn to map the surrounding environment directly to safe trajectory points that avoid obstacles.

The core of our approach is the PPO algorithm, where the clipping method plays a critical role. The clip parameter significantly influences the performance of the algorithm. During training, we experimented with various clip parameter values. Fig. 5 shows the impact of different clip parameters on reward convergence. Quantitatively, with $\epsilon = 0.01$, the algorithm achieves a final reward of 3200 with consistent stability, while $\epsilon = 0.1$ reaches 3038 but shows higher variance. The $\epsilon = 0.01$ configuration de-

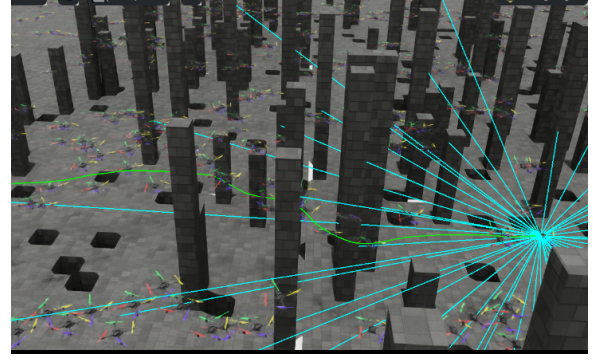


Fig. 4: RL training and evaluation environment. The green line represents the flight path of the UAV, and the blue line represents the lidar ray from the UAV.

livers 5% higher rewards with lower variance. Therefore, we selected $\epsilon = 0.01$ for our experiments.

The PPO algorithm is primarily used to generate safe flight points in real-time while ensuring efficient path planning with obstacle avoidance capabilities. Figs. 6 and 7 illustrate the path length and safety metrics corresponding to the policies obtained after different training iterations in evaluation mode. With $\epsilon = 0.01$, the path length decreases from 48m to 23.4m (with the direct distance being 20m), indicating that the UAV progressively learns to navigate around obstacles while minimizing detours. Meanwhile, the safety metric improves dramatically from 30 to 997.6, demonstrating the algorithm's significantly enhanced ability to maintain safe distances from obstacles during flight. This substantial improvement in safety metrics directly reflects the development of effective obstacle avoidance behavior through our reinforcement learning approach. The $\epsilon = 0.1$ configuration achieves a path length of 24.8m and safety value of 946.9, which are 5.6% longer and 5.4% lower respectively. These results demonstrate that the proposed strategy with $\epsilon = 0.01$ exhibits stronger convergence properties and delivers better performance in both path optimization and obstacle avoidance.

The training process involves the UAV learning to balance the competing objectives of minimizing path length and avoiding obstacles. As training progresses, the policy evolves from initially prioritizing shorter paths (often at the expense of safety) to developing sophisticated navigation strategies that effectively detect and circumvent obstacles while maintaining efficient trajectories. The steady improvement in both metrics confirms that our approach successfully learns obstacle avoidance behaviors without requiring explicit programming of avoidance maneuvers.

D. Ablation Experiment

1. Cluster Sequence Allocation for Multi-UAV

This experiment focuses on evaluating the results of cluster sequence allocation for multi-UAV systems. Our study proposes a method based on the transformation of

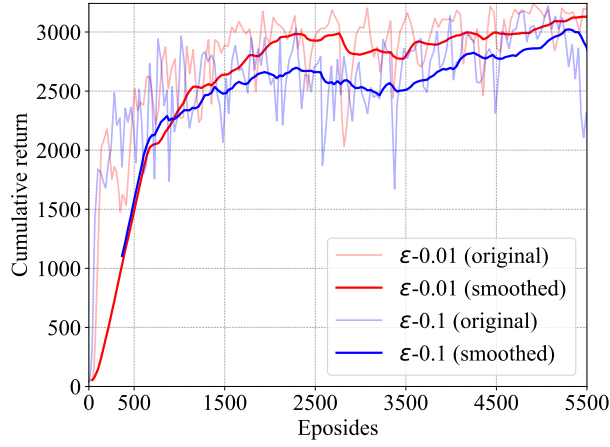


Fig. 5: RL reward convergence

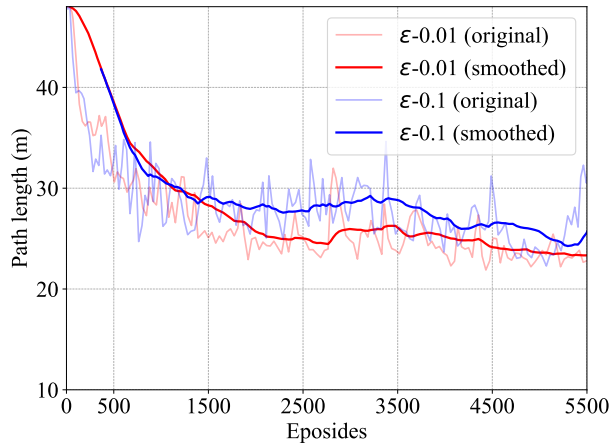


Fig. 6: Path length during evaluation

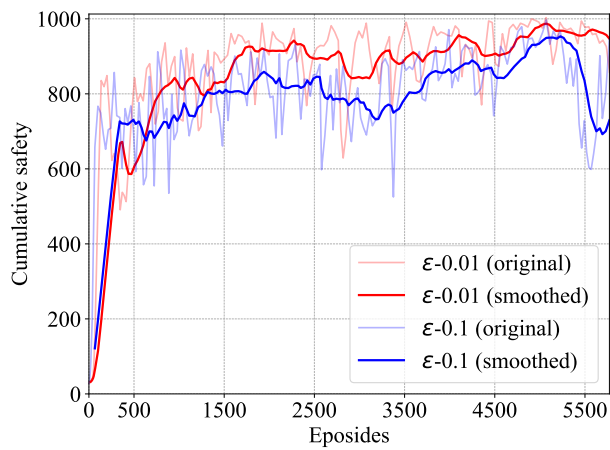


Fig. 7: Safety during evaluation

the mTSP (multi-Traveling Salesman Problem) to address this issue, wherein each cluster is positioned directly above its corresponding cluster head node. Fig. 8 presents the path lengths associated with cluster allocation and sequence planning across different algorithms. The parameters and experimental setup are consistent with those

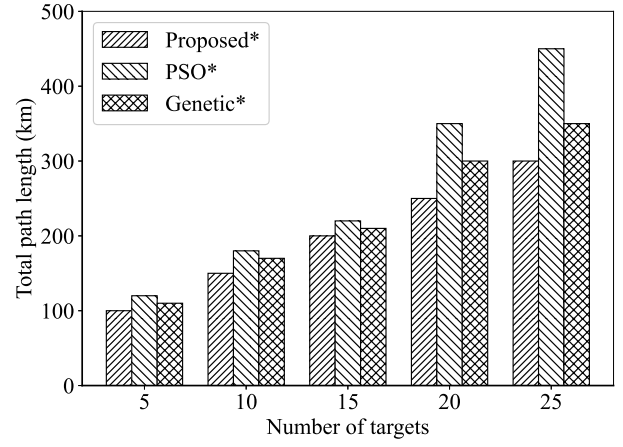


Fig. 8: Transformation Method for Cluster Allocation

in Section VI.B. For 25 clusters, our method achieves a total path length of 300m, while PSO and Genetic algorithms result in 450m and 350m respectively, representing improvements of 33.3% and 14.3%. Similar performance advantages are observed across all cluster configurations (5-20 clusters), with our method consistently achieving 10-33% shorter paths. These results demonstrate that the proposed allocation strategy delivers superior performance in minimizing total path lengths for cluster visits.

2. Dynamic Programming for Hovering Points Selection

In this study, the dynamic programming algorithm is employed to fine-tune the selection of hovering points, balancing energy consumption during communication with that during flight. This ablation experiment evaluates the total energy consumption, flight energy consumption, and communication energy consumption before and after the fine-tuning of hovering points. As shown in Fig. 9, for an identical cluster sequence, the hovering points sequence derived from the dynamic programming algorithm proposed in this study achieves the lowest total energy consumption. For 25 IoT nodes, our method achieves 141.14 Wh total energy, saving 3.3% to 6.1% compared to other methods. Furthermore, Fig. 10 demonstrates that, compared to the fixed selection strategy where the hovering point is positioned directly above the cluster head node, the proposed algorithm identifies the set of hovering points that minimizes flight energy consumption, reducing flight energy by 2.2% to 5.4%. Conversely, as anticipated, Fig. 11 shows that the proposed algorithm results in increased communication energy consumption relative to the fixed selection strategy, though it remains 10.0% to 19.5% lower than PSO and Genetic algorithms. This confirms Remark 2 that hovering directly above cluster heads minimizes communication energy. However, the results in Fig. 10 and Fig. 11 clearly show that the overall flight energy consumption is approximately 6-8 times greater than the energy used for hovering and communication, thereby supporting our statement in

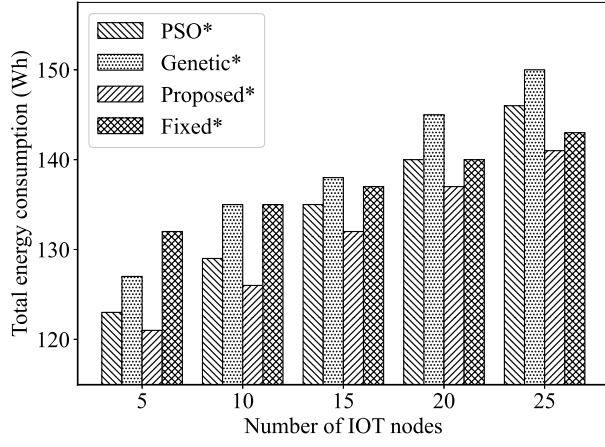


Fig. 9: Total energy for given cluster sequence

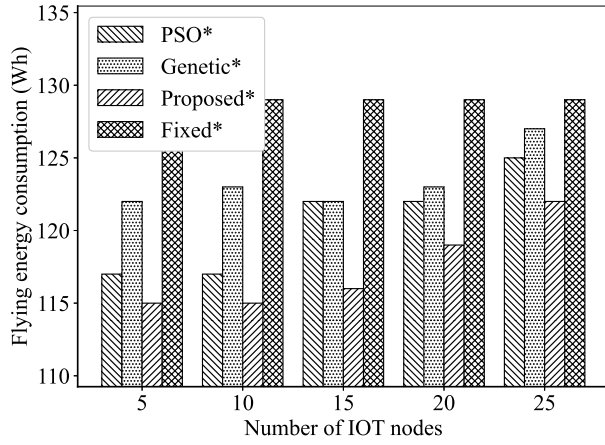


Fig. 10: Flying energy for given cluster sequence

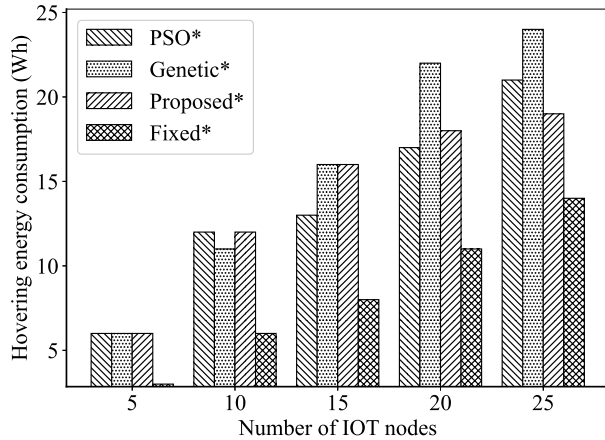


Fig. 11: Communication energy for given cluster sequence

Remark 1 and confirming the effectiveness of our dynamic programming approach.

3. Trajectory Optimization of Given Safe Flight Points

In our study, UAVs operating in complex static and dynamic environments identify safe flight points in real-

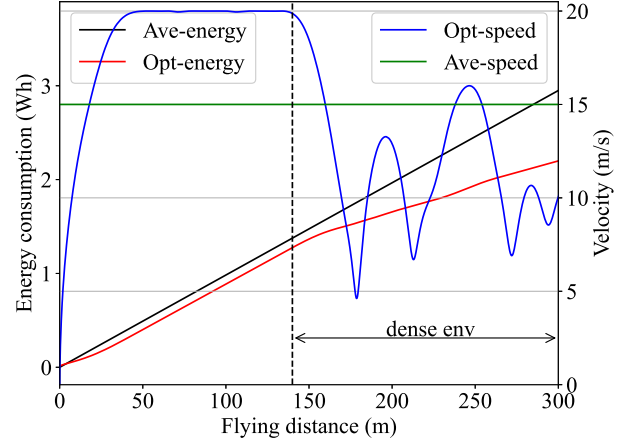


Fig. 12: Flight energy for given safe flight points

time using RL algorithms. Subsequently, trajectory optimization algorithms generate control commands to be executed by the UAV's low-level controllers. Most existing studies [10] assume that UAVs maintain a constant speed between trajectory points. To challenge this assumption, we compare energy consumption between constant-speed flight and our optimized variable-speed flight trajectories in this experiment.

As illustrated in Fig. 12, we evaluated energy consumption for the same flight path length using both our optimized flight strategy and the constant-speed strategy. In this ablation experiment, the constant-speed flight was conducted at an average speed determined by the method in [31], which identifies the speed that maximizes flight distance for a given total energy. For our parameters, this speed is 15 m/s, with the UAV's maximum speed in the test environment set to 20 m/s. Notably, at the 140-meter mark, the UAV enters a dense obstacle environment, where its speed begins to decrease significantly.

The results indicate that for the same flight distance, our optimized flight strategy enables the UAV to fly with lower energy consumption. It is particularly evident in dense obstacle environments, where our trajectory optimization strategy demonstrates a clear advantage in energy efficiency over the constant-speed approach.

VI. Conclusions

This paper presents a comprehensive approach to the joint optimization of task allocation and path planning for multi-UAV systems in a cluster-based IoT environment. By decomposing the complex problem into subproblems, such as cluster sequence allocation, hovering point selection, safe path planning, and trajectory optimization, our method significantly reduces UAV energy consumption while ensuring safe and efficient data collection. Simulation results validate the effectiveness of our approach, demonstrating superior performance in dynamic environments compared to traditional methods. Future work will be focused on extending the framework

to larger and more complex IoT networks, incorporating real-world constraints, and further enhancing real-time decision-making capabilities of UAVs.

APPENDIX

Let the total energy consumption model for the UAV be given by

$$E_{\text{total}} = \int_0^T (P_{\text{fly}}(t) + P_{\text{comm}}(t)) dt,$$

where $P_{\text{fly}}(t)$ is the power consumed for flight (e.g., propulsion), $P_{\text{comm}}(t)$ is the power consumed for communication, and T is the flight time. The optimization problem typically aims to minimize E_{total} subject to various constraints.

Now consider that the computational power P_{comp} required for autonomous navigation is determined by the frequency of the navigation module, and since this frequency is fixed, P_{comp} can be treated as a constant. With the addition of this constant power, the total energy consumption becomes

$$E'_{\text{total}} = \int_0^T (P_{\text{fly}}(t) + P_{\text{comm}}(t) + P_{\text{comp}}) dt.$$

By the linearity of integration, this expression can be split into

$$E'_{\text{total}} = \int_0^T (P_{\text{fly}}(t) + P_{\text{comm}}(t)) dt + \int_0^T P_{\text{comp}} dt.$$

Since P_{comp} is constant,

$$\int_0^T P_{\text{comp}} dt = P_{\text{comp}} \cdot T.$$

Thus, the total energy becomes

$$E'_{\text{total}} = E_{\text{total}} + P_{\text{comp}} \cdot T.$$

Now, let the original optimization problem be formulated as

$$\min_{\mathbf{u}} E_{\text{total}} = \min_{\mathbf{u}} \int_0^T (P_{\text{fly}}(t) + P_{\text{comm}}(t)) dt,$$

where $\mathbf{u}$ represents the decision variables (e.g., flight trajectory, speed, communication resource allocation, etc.). With the addition of the constant computational power term, the new objective function becomes

$$J'(\mathbf{u}) = J(\mathbf{u}) + P_{\text{comp}} \cdot T,$$

where

$$J(\mathbf{u}) = \int_0^T (P_{\text{fly}}(t) + P_{\text{comm}}(t)) dt.$$

Consider two cases:

Case 1: If T does not depend on the decision variables $\mathbf{u}$, then $P_{\text{comp}} \cdot T$ is a constant term. In this case, adding a term that is independent of $\mathbf{u}$ does not affect the optimal solution, since

$$\nabla_{\mathbf{u}} J'(\mathbf{u}) = \nabla_{\mathbf{u}} J(\mathbf{u}) + \nabla_{\mathbf{u}} (P_{\text{comp}} \cdot T) = \nabla_{\mathbf{u}} J(\mathbf{u}).$$

Thus, the original optimality condition $\nabla_{\mathbf{u}} J(\mathbf{u}) = \mathbf{0}$ remains unchanged.

Case 2: If T is a decision variable, note that since P_{comp} is constant and the term $P_{\text{comp}} \cdot T$ appears linearly, its first derivative is constant. In the necessary optimality conditions (such as the Euler-Lagrange equations), adding this linear term results in a constant offset that does not alter the optimality conditions determined by the other nonlinear terms. Therefore, the optimal values of the decision variables remain unchanged.

In conclusion, regardless of whether T is a decision variable, adding the constant computational power term yields

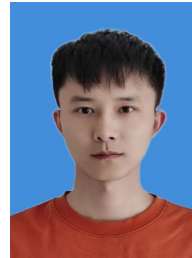
$$E'_{\text{total}} = E_{\text{total}} + P_{\text{comp}} \cdot T.$$

Since P_{comp} is constant, this additional term does not affect the optimization problem (or only adds a constant shift) and therefore does not change the optimal solution of the original optimization problem (Eq. 30).

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Guodong Zhao (Student Member, IEEE) received his B.S. degree in Information and computational science from the Shandong University of Science and Technology, Qingdao, China in 2018, and the M.S. degree in computer system architecture from Northeastern University, Shenyang, China in 2021. From 2021 to 2023, he worked as a senior R&D engineer for AI high performance computing at Baidu. He is currently pursuing the Ph.D. degree in aircraft design from Beihang University, Beijing, China. His research interests lie in the area of UAV and AI.



Jingjing Wang (IEEE Senior Member) received his B.Sc. degree in electronic information engineering from Dalian University of Technology, Liaoning, China in 2014 and the Ph.D. degree in information and communication engineering from Tsinghua University, Beijing, China in 2019, both with the highest honors. From 2017 to 2018, he visited the next generation wireless group chaired by Prof. Lajos Hanzo from University of Southampton, UK. Dr. Wang is currently a professor at the School of Cyber Science and Technology, Beihang University, Beijing, China, and also a researcher at Hangzhou Innovation Institute, Beihang University, Hangzhou, China. His research interests include AI enhanced next-generation wireless networks, UAV networking and swarm intelligence. He has published over 100 IEEE Journal/Conference papers. He is currently serving as an Editor for the IEEE Transactions on Vehicular Technology, IEEE Internet of Things Journal and IEEE Wireless Communications Letter. Dr. Wang was a recipient of the Best Journal Paper Award of IEEE ComSoc Technical Committee on Green Communications & Computing, the Best Paper Award of the IEEE ICC and the IEEE IWCMC.



Zhijun Meng received the Ph.D. degree from the School of Aeronautic Science and Engineering, Beihang University, China in 2009. He is currently a professor in aerospace science and engineering with Beihang University. His research interests lie in the area of autonomous flight and collaborative exploration of UAV.



Zichen Wang received his B.S. degree in Vehicle Engineering from the Beijing Institute of Technology, Beijing, China in 2020. He is currently pursuing the Ph.D. degree in the School of Aeronautic Science and Engineering from Beihang University, Beijing, China. His research interests lie in the area of UAV path planning and autonomous exploration.



Hang Fu (Student Member, IEEE) received his B.S. degree in flight vehicle design & engineering from the Northwestern Polytechnical University, Xi'an, China in 2020, and the M.S. degree in mechanical engineering from Beihang University, Beijing, China in 2023. He is currently pursuing the Ph.D. degree in the School of Cyber Science and Technology from Beihang University, Beijing, China. His research interests lie in the area of low-altitude intelligent networks, and multi-agent system.



Chunxiao Jiang (Fellow, IEEE) is an associate professor in School of Information Science and Technology, Tsinghua University. He received the B.S. degree in information engineering from Beihang University, Beijing in 2008 and the Ph.D. degree in electronic engineering from Tsinghua University, Beijing in 2013, both with the highest honors. From 2011 to 2012 (as a Joint Ph.D) and 2013 to 2016 (as a Postdoc), he was in the Department of Electrical and

Computer Engineering at University of Maryland College Park under the supervision of Prof. K. J. Ray Liu. His research interests include application of game theory, optimization, and statistical theories to communication, networking, and resource allocation problems, in particular space networks and heterogeneous networks. Dr. Jiang has served as an Editor of IEEE Transactions on Communications, IEEE Internet of Things Journal, IEEE Wireless Communications, IEEE Transactions on Network Science and Engineering, IEEE Network, IEEE Communications Letters, and a Guest Editor of IEEE Communications Magazine, IEEE Transactions on Network Science and Engineering and IEEE Transactions on Cognitive Communications and Networking. He has also served as a member of the technical program committee as well as the Symposium Chair for a number of international conferences. Dr. Jiang is the recipient of the Best Paper Award from IEEE GLOBECOM in 2013, IEEE Communications Society Young Author Best Paper Award in 2017, the Best Paper Award from ICC 2019, IEEE VTS Early Career Award 2020, IEEE ComSoc Asia-Pacific Best Young Researcher Award 2020, IEEE VTS Distinguished Lecturer 2021, and IEEE ComSoc Best Young Professional Award in Academia 2021. He received the Chinese National Second Prize in Technical Inventions Award in 2018 and Natural Science Foundation of China Excellent Young Scientists Fund Award in 2019.